Project Purpose

What are we building?

The goal of this project is to build a streamlined group of assignments, which trains undergraduates and recent hires to carry out data science in the Computational Memory Lab.

What are the deliverables?

1. A sequenced set of hands-on data science assignments
2. A set of assignment introductions which effectively teaches our lab's research

What are the benefits?

1. New undergrads and hires ramp up faster with less ambiguity
2. Fewer repetitive training meetings / repeated explanations from senior staff
3. More consistent skill level across trainees
4. Creates a shared “lab standard” for how data science work is done

What are our metrics of success?

1. Increased trainee independence
2. Increased percentage of trainees completing the training
3. Decreased reliance on senior lab members to teach.

All Edits

**all BIDS edits are highlighted*

Added bidsreader as submodule in dependencies/bidsreader

Introduction 0 (Python Numpy Pandas)

- Added introductory paragraphs clarify the specific goals of Introduction and Assignment 0
- Added/Expanded sections on handling errors and print debugging
- Certain sections and exercises made optional depending on relevancy to student

Assignment 0 (Python Numpy Pandas)

- Disclaimer statement added at the top explaining the need for personal research for inexperienced coders.
- Problem 1
    - Completely Removed
- Problem 2
    - Part C is shortened and additional direction in terms of how to graph the data is included.
- Problem 4
    - Removed part of the question using pickle as it is not yet covered in introductions

Added: Introduction 1A (OpenBIDS)

- Section 0: Introduce OpenBIDS Data Structure

- Section 1: Loading Scalp BIDS Data (Events and EEG)
- Section 2: Load RAM BIDS iEEG data (Monopolar and Bipolar) and Electrode, Coordsystem, and Channel data.

Introduction 1 (Behavioral Analysis of Memory)

- Renamed to "Behavioral Analysis With BIDSReader"
- Got rid of working with data index and replace it with BIDSReader's get metadata functions
    - Define subject, experiment, task at top of page instead of deriving it from data index
- Added section to teach people how to load events from beh folder and eeg folder
- Basic restructuring and writing edits to make sections easier to understand. Clarifications/expansions include:
    - Data Frame Columns
    - Scalp vs. EEG data
    - Events Data Frame
- Practice exercises made optional

Assignment 1 (Behavioral Analysis of Memory)

- Introduction
    - Added section on how to download data and how to set up a local BIDS database.
    - Moved the import section to Introduction.
- Question 1
    - Hint added to help locate Jefferson Hospital Data
- Question 2
    - Modified to focus on recall percentages instead of IRTs
- Question 3
    - Recommendation made to reference FHM in explanation of the Serial Position Curve. The explanation as a whole is tweaked to make things clearer to the student
- Question 4
    - Reduced to no distractor and 16000ms graphs. The same point can be communicated quicker.
    - No longer import cmlreaders.
    - Load events using MNE-BIDS.

Introduction 2 (IRT - Lag CRP - PLI)

- "Print Debugging" and "Handling Errors" moved to introduction 0
- Substantial expansion of following sections
    - IRT Curves
    - Lag-CRP
    - PLI Recency

Assignment 2 (IRT - Lag CRP - PLI)

- Minimal Edits

- Added mne_bids imports and how to download data section to top.
- Added subject_sessions_with_task function to find all subjects and sessions related to a task.
    - In Assignment Overview, load events using bids.

Introduction 3 (EEG and ERPs)

- Certain sections marked optional based on relevance to the student
- Expansion/clarification of following topics:
    - Pairs vs. Contacts
    - PTSA vs. MNE
    - EEG timeseries
    - Bipolar Referencing
    - Subsequent Memory Effect
- Added description after “Pairs vs. Contacts” how to load electrodes and how BIDSReader automatically infers coordinate system
- Rewrite Key Regions section which loads the electrode metadata and channel metadata. We also load the merged channel and electrode dataframes
- Convert to PTSA is now after loading EEG data
- Added code to calculate recalled data

Assignment 3 (EEG and ERPs)

- Questions 3, 4, and 5 have been eliminated
- Revised Q1
- New Q2
- Added link to data.

Introduction 4 (Statistics)

- Added section on confidence intervals

Assignment 4 (EEGs and ERPs)

- Eliminated entirely.
- Statistics involved with this assignment are redistributed to Assignment 3.

Introduction 5 (Spectral Analysis)

- Several sections made optional
- Expansion/clarification of following topics
    - Wavelets
    - Tips for Spectral EEG
    - PSTA/MNE
    - Improved explanations tying all graphs to key scientific concepts
- Convert CMLReader to BIDSReader

Assignment 5 (Spectral Analysis)

- Question 2 removed
- Added link to data.

Introduction 6
- Brief beginner friendly introduction added
- Several sections made optional
- Brief explanations added to explain the conceptual importance of certain code blocks
- Modifications to following sections with the goal of being more beginner friendly
    - Recommended Best Practices
    - Parallel Computing Tips

Assignment 6
- Brief explanation added in question 1
- Assignment largely kept the same
- Provided additional clarification for question 3

Introduction 7
- Introduction is made optional
- Convert CMLReader to BIDSReader

Assignment 7
- Remove question 1.2

Introduction 8
- Changed to introduction 6B to increase course continuity

Assignment 8
- Kept entirely the same

Introduction 11
- Convert CMLReader to BIDSReader

Introduction 12
- Convert CMLReader to BIDSReader